

Mechanical Power Module (MPM) — Logistics Specification v1.0

Microsoft-Aligned Standards Edition

Takeaway: The Logistics Specification defines the **complete movement, storage, and deployment contract** for MPM units: container geometry, palletization, handling requirements, tracking metadata, environmental storage limits, and field-deployment workflows. It ensures that MPM modules can be shipped, warehoused, and deployed safely and predictably across manufacturers, sovereigns, and hyperscalers.

1. Logistics Model Overview

The MPM logistics model consists of five coordinated layers:

- **Containerization**
- **Palletization & Handling**
- **Storage Requirements**
- **Deployment Workflow**
- **Tracking & Metadata**

Each layer is mandatory for certification.

2. Containerization Specification

MPM units ship in a universal protective container designed for global logistics.

2.1 Container Geometry

- External dimensions: **600 mm × 400 mm × 350 mm**
- Internal protective cavity: **500 mm × 300 mm × 300 mm**
- Weight (empty): **4.5 kg**
- Material: High-impact polymer with steel reinforcement
- Stackability: Up to **6 units** high
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2.2 Shock & Vibration Protection

- Shock absorption: **≥ 30 g**
- Vibration damping: **≥ 60% reduction**
- Internal foam density: **45 kg/m³**

2.3 Environmental Protection

- IP rating: **IP54 minimum**
- UV-resistant outer shell
- Corrosion-resistant hardware

2.4 Sealing & Security

- Dual-latch locking system
- Tamper-evident seal compatible
- RFID tag mount point

3. Palletization & Handling

Defines how MPM units are grouped and moved through industrial logistics.

3.1 Pallet Configuration

- Standard pallet: **1200 mm × 1000 mm**
- Units per pallet: **8**
- Max pallet weight: **≤ 120 kg**

3.2 Tie-Down Requirements

- Minimum of **4 straps**
- Strap tension: **≥ 250 N**
- No strap may compress the safety shroud

3.3 Forklift Handling

- Fork entry height: **≥ 90 mm**
- No fork may contact container walls
- Lift from pallet only, never individual containers

3.4 Manual Handling

- Container handles rated for **200 N**
- Two-person lift recommended
- No manual lift permitted above **25 kg**

4. Storage Requirements

Defines environmental and positional requirements for safe storage.

4.1 Temperature

- Long-term storage: **-10°C to +40°C**
- Short-term exposure: **-20°C to +50°C**

4.2 Humidity

- Non-condensing
- Relative humidity: **≤ 90%**

4.3 Orientation

- Must remain upright
- Tilt tolerance: **≤ 10°**

4.4 Shelf Storage

- Shelf load rating: **≥ 150 kg**
- Minimum clearance above unit: **50 mm**

4.5 Long-Term Storage Interval

- Inspection required every **12 months**
- Replace desiccant every **24 months**

5. Deployment Workflow

Defines the standardized field-deployment process for MPM units.

5.1 Pre-Deployment Inspection

- Container integrity
- Tamper seal status
- RFID tag readability
- Visual inspection of shroud and frame

5.2 Site Preparation

- Clear Zone 2 safety envelope
- Confirm mounting frame compatibility

- Verify controller interface availability

5.3 Deployment Sequence

1. Remove unit from container
2. Inspect mechanical interface
3. Align using Datum C
4. Engage lock mechanism
5. Initiate controller handshake
6. Perform safety interlock check
7. Begin operation

5.4 Post-Deployment Logging

- Deployment timestamp
- Operator ID
- Location metadata
- Initial telemetry snapshot
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6. Tracking & Metadata

Defines the universal tracking schema for MPM units.

6.1 Required Identifiers

- MPM Serial Number
- Manufacturing Batch ID
- Compliance Certificate ID
- RFID Tag ID
- Optional: QR code

6.2 Metadata Schema

Field	Description
Serial Number	Unique module identifier
Manufacture Date	ISO-8601 format

Field	Description
Compliance Level	Base / Enhanced / Derivative
Hazard Envelope Class	A / B / C
Controller Protocol Version	v1.0+

6.3 Digital Tracking

- RFID readable at ≥ 1 m
- Optional Bluetooth beacon
- Cloud registry integration supported
- Telemetry snapshot stored at each deployment

6.4 Chain-of-Custody

- Required for all certified units
- Must include:
 - Transfer timestamp
 - Origin and destination
 - Responsible operator
 - Condition report

7. Logistics Compliance Requirements

A module is logistics-certified only if it:

- Fits the universal container geometry
- Passes shock, vibration, and environmental tests
- Meets palletization and handling rules
- Supports full metadata schema
- Maintains chain-of-custody integrity
- Passes all logistics tests in the **Compliance Test Suite**

8. Logistics Test Summary Table

Test	Objective	Pass Criteria
Container Drop Test	Verify shock protection	No internal damage
Pallet Stability Test	Validate pallet load	No tipping or shifting
RFID Readability Test	Verify tracking	Readable ≥ 1 m

Test	Objective	Pass Criteria
Environmental Storage Test	Validate storage limits	No degradation
Deployment Workflow Test	Validate field deployment	All steps pass